

IB MATH AA SL

Lesson 2

Arithmetic Sequences & Series

Number & Algebra · 60-minute lesson

Syllabus SL 1.2 · Property of Lynda Badmus Education



Do Now — recall

ON YOUR WHITEBOARD

1. Write the next two terms of 5, 8, 11, 14, ...
2. What is being added each time?
3. In 20, 17, 14, ... what is the common difference?

Recall:

term-to-term: add the same number

Answers (click to reveal)

1. 17, 20
2. +3 each time
3. $d = -3$ (it decreases)

What we're learning today

LEARNING OBJECTIVES

- 1** **Recognise**
an arithmetic sequence and find the common difference d .
- 2** **Apply**
 $u_n = u_1 + (n-1)d$ to find any term.
- 3** **Apply**
 $S_n = n/2 (2u_1 + (n-1)d)$ to sum a series.
- 4** **Solve**
problems where two pieces of information give u_1 and d .

SUCCESS CRITERIA — I CAN...

- ✓
find d by subtracting consecutive terms
- ✓
find any term with the n th-term formula
- ✓
find the sum of the first n terms
- ✓
set up and solve simultaneous equations for u_1 and d

The n th term

THE FORMULA

$$u_n = u_1 + (n-1)d$$

u_1

first term

d

common difference = $u_{n+1} - u_n$

n

term number

SPOTTING d

Subtract any term from the one after it.

2, 5, 8, 11 $\rightarrow d = 3$

10, 7, 4 $\rightarrow d = -3$

Care:

d can be negative or a fraction.

Find a term

QUESTION

An arithmetic sequence has $u_1 = 7$ and $d = 4$.

Find the 10th term.

SOLUTION

Step 1 $u_1 = 7, d = 4, n = 10$

Step 2 substitute into the formula:

$$u_{10} = 7 + (10 - 1)(4)$$

Step 3 evaluate:

$$u_{10} = 7 + 36 = 43$$

Tip:

Write the formula and the substitution line first — that earns the method mark.

Your turn — nth term

ANSWER IN YOUR BOOK

1. Find the 8th term of 3, 7, 11, ...
2. An AP has $u_1 = 20$, $d = -3$. Find u_6 .
3. Find d if $u_1 = 5$ and $u_4 = 23$.

Answers (click to reveal)

1. $3+7(4)=31$
2. $20+5(-3)=5$
3. $23=5+3d \rightarrow d=6$

The sum of n terms

THE SUM FORMULA

$$S_n = \frac{n}{2} (2u_1 + (n-1)d)$$

Use when you know u_1 and d .

Also

$S_n = n/2 (u_1 + u_n)$ if you know the last term.

WHEN TO USE WHICH

Know u_1 and $d \rightarrow$ first form.

Know first and last term $\rightarrow$ second form.

Both give the same answer.

Sum a series

QUESTION

Find the sum of the first 10 terms of

$$7 + 11 + 15 + \dots$$

SOLUTION

Step 1 $u_1 = 7, d = 4, n = 10$

Step 2 substitute:

$$S_{10} = \frac{10}{2} (2(7) + (10 - 1)(4))$$

Step 3 evaluate:

$$S_{10} = 5(14 + 36) = 250$$

Your turn — sums

ANSWER IN YOUR BOOK

1. Sum the first 12 terms of $2 + 5 + 8 + \dots$
2. An AP has $u_1 = 50$, $d = -4$. Find S_{10} .
3. Sum the first 20 terms of $1 + 3 + 5 + \dots$

Answers (click to reveal)

1. $12/2(2+11\cdot3)=222$
2. $10/2(100+9\cdot-4)=320$
3. $20/2(2+19\cdot2)=400$

Exam-style question

An arithmetic sequence has third term 17 and seventh term 33.

- (a) Find the common difference d . [2]
- (b) Find the first term u_1 . [2]
- (c) Find the sum of the first 15 terms. [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $u_7 - u_3 = 4d = 33 - 17 = 16$ (M1) $\rightarrow d = 4$ (A1)
- (b) $u_1 + 2(4) = 17$ (M1) $\rightarrow u_1 = 9$ (A1)
- (c) $S_{15} = 15/2 (2 \cdot 9 + 14 \cdot 4)$ (M1)(A1) $= 615$ (A1)

Common mistakes



d, not the term

d is what you ADD, not the first term.



(n-1), not n

$u_n = u_1 + (n-1)d$ — the multiplier is (n-1).



Sign of d

If the sequence decreases, d is negative.



Sum vs term

S_n adds terms; u_n is a single term — don't mix the formulas.

Mixed practice

QUESTIONS

1. Find the 15th term of 4, 9, 14, ...
2. An AP has $u_1 = 100$, $d = -7$. Find u_{10} .
3. Sum the first 25 terms of $3 + 7 + 11 + \dots$
4. The 2nd term is 11 and the 5th is 26. Find u_1 and d .

Answers (click to reveal)

1. $4 + 14 \cdot 5 = 74$
2. $100 + 9 \cdot -7 = 37$
3. $25/2(6 + 24 \cdot 4) = 1275$
4. $3d = 15 \rightarrow d = 5$; $u_1 = 11 - 5 = 6$

Plenary & exit ticket

KEY FORMULAS

$$u_n = u_1 + (n - 1)d$$

$$S_n = \frac{n}{2}(2u_1 + (n - 1)d)$$

EXIT TICKET — before you leave

1. Find the 6th term of 2, 9, 16, ...
2. Find d for 40, 34, 28, ...
3. Sum the first 10 terms of 5 + 8 + 11 + ...

Answers (click to reveal)

1. $2 + 5 \cdot 7 = 37$
2. $d = -6$
3. $10/2(10 + 9 \cdot 3) = 185$

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Arithmetic sequences & series.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 3

Geometric sequences & series — when each term is MULTIPLIED by a fixed ratio.

$$u_n = u_1 r^{n-1}$$